

Forced to face the truth: A meta-analysis on the effectiveness of moral reminders



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The room for pro-social policy interventions

Individuals, when taking a selfish decision, may distort or selectively ignore information to preserve their self-image (Bénabou and Henkel, 2025). To do so, they may manipulate:

- the perceived selfishness of their own current actions (Vu et al. (2023), Exley (2020));
- or of their past actions (Saucet and Villeval, 2019);
- or the norms with which they are willing to comply (Bicchieri et al., 2023).

Moral reminders (moral appeals, moral nudges ... “Do the right thing” kind of interventions), widely used by policy makers, firms, NGOs, can be used in order to **counteract such manipulations**.

THIS PAPER: Comprehensive meta-analytical review on the efficacy of “moral reminders”.

What we do

- Collect comprehensive data on 415 experiments evaluating the impact of moral reminders on pro-social behaviours, intentions or attitudes.
- Document significant heterogeneity across a wide range of disciplines (economics, psychology, business, social sciences...), methodologies (experimental games in the lab, in the field...) or over time (1982-2020).
- Identify the average treatment effect of moral reminders, as well as heterogeneity in their effectiveness.
- And last but not least, propose a conceptual framework for “moral reminders”.

Conceptual framework: a classification of “moral reminders”

- **Priming:** exposure to a task (word decoding, text reading) aimed at making salient the ethical drivers of a decision (Dai et al., 2023).
 - *“Participants had to write words from a list of positive traits words such as caring, generous, fair, and kind and then had to write a story using those words”* (Sachdeva et al., 2009, dictator game).
- **Information on consequences:** about the consequences of the available choices;
 - *“Your tax payment contributes to the funding of publicly financed services in education, health and other important sectors of society”* (Bott et al., 2020).
- **Information on descriptive norms:** information about the behavior of a group of individuals (what others do);
 - *“The vast majority of participants were honest about the reported number in the dice task”* (Dimant et al., 2020, dice-rolling task).
- **Information on injunctive norms:** information about what people should or should not do (what others think).
 - *“The vast majority of participants said that one should be honest about reported number in the dice task.”* (ibid.).

Open questions

- How effective are moral reminders?
- What variation according to studies design?
- In what context do they work best? When do they backfire?
- What kind of moral reminders are most effective?

Hence contributing to a growing literature assessing the policy-relevance of behaviorally informed interventions:

- Efficacy of choice architecture interventions ([Mertens et al., 2022](#)), commitment devices (e.g., oaths; [Zickfeld et al., 2024](#)), observability ([Bradley et al., 2018](#)), behavioral climate interventions ([Vlasceanu et al., 2024](#)), tax compliance ([Antinyan and Asatryan, 2024](#)), or (even) meditation ([Kreplin et al., 2018](#)).
- The scalability of nudge interventions ([DellaVigna and Linos, 2022](#));
- Complementarity with price incentives ([Chetty et al., 2014](#));
- Cost-effectiveness of nudges vs traditional policies ([Benartzi et al., 2017](#)).

Data

Systematic review and data collection

- ① **Keywords and starting sample** defined from references and secondary references of relevant papers from the literature ([full list](#));
- ② **Systematic review** using [Rayyan](#), i.e., keyword analysis of the titles and abstract of references available on [Google scholar](#) (2500+ refs), **until 2020**.
- ③ **Inclusion decisions** on all references rated 3+ stars (excluding in particular non-experimental evidence).
⇒ 88 research articles ([full list](#)),
- ④ **Individual-level data collection** (DataStudies, DS; $K_{\text{DS}} = 168$) from public sources (68 obs.) or by contacting the authors (68% response rate).
- ⑤ **Data consolidation.** Retrieve missing info from DS (113) so as to measure effect sizes on 241 observations overall; 174 can be further included by making assumptions about missing info ([more info](#)).
- ⑥ $K = 415$ **treatment effects**, from 88 papers featuring $n = 115$ different studies (excluding 31 studies with a selfish reminder; see [distribution](#)).

Dataset: studies overview

	Priming	Consequences	Descriptive norm	Injunctive norm	Overall
	21.0%	14.5%	32.5%	32.0%	100%
Discipline (%)					
Business	9.2	3.3	5.2	0.0	4.1
Economics	14.9	26.7	27.4	49.6	31.8
Psychology	70.1	40.0	62.2	37.6	52.8
Social Sciences	1.1	15.0	3.0	8.3	6.0
Others	4.6	15.0	2.2	4.5	5.3
Country (% , missing otherwise)					
Asia	2.3	6.7	0.0	8.3	4.1
Europe	31.0	18.3	41.5	20.3	29.2
North-America & Australia	64.4	56.7	54.1	66.9	60.7
Rest of the world	1.1	0.0	0.0	0.8	0.5
Missing	1.1	18.3	4.4	3.8	5.5
Year of publication (%)					
1982	0.0	0.0	0.0	1.5	0.5
1990's	0.0	13.3	8.9	13.1	9.0
2000-2005	10.3	3.3	1.5	0.8	3.4
2006-2010	16.1	6.7	5.2	2.3	6.8
2011-2015	35.6	28.3	43.0	50.0	41.5
2016-2020	37.9	48.3	41.5	32.3	38.8
Measure of the outcome					
Experimental game	60.9	13.3	32.6	44.4	39.5
Field task	11.5	53.3	45.2	37.6	36.9
Scale	27.6	33.3	22.2	18.0	23.6

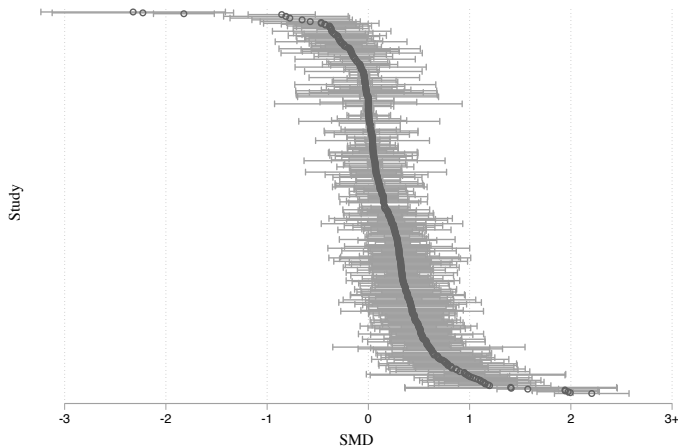
Distribution of sample size

Results:

Overall distribution of the effect

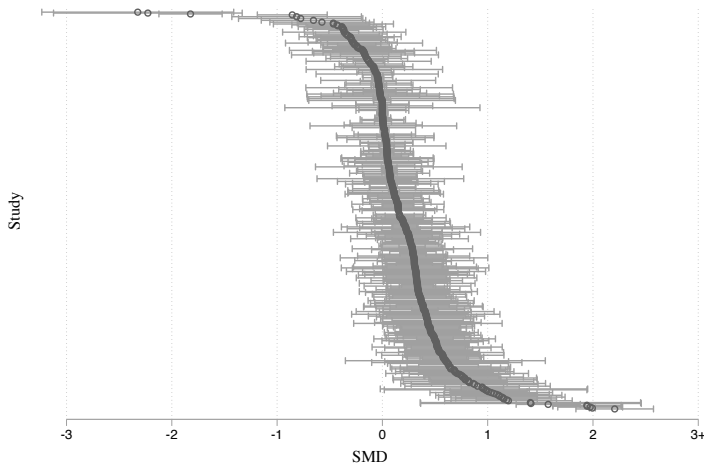
How effective are MR according to included studies? (excluding outliers; also see Galbraith plot)

Density



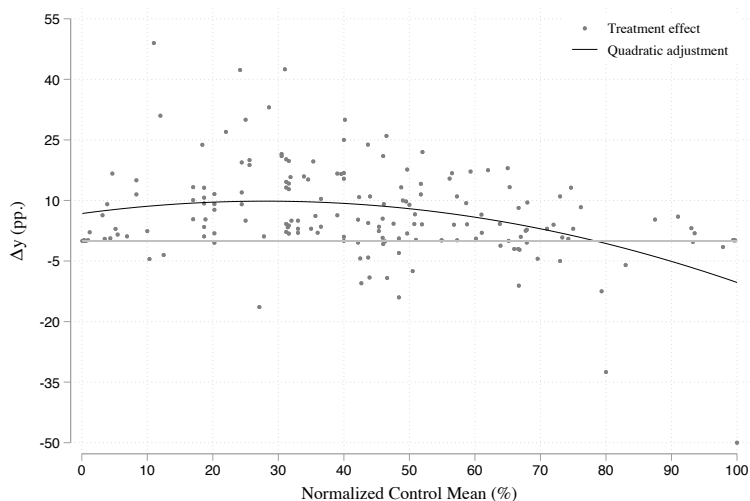
Average effect (RE model): $\widehat{SMD} = 0.241 \in [0.179; 0.303], p = 0.000$.

vs, e.g., 0.430 in [Mertens et al. \(2022, pro-environmental nudges\)](#);
0.238 in [Antinyan and Asatryan \(2024, tax-compliance nudges\)](#);
0.270 in [Zickfeld et al. \(2024, commitment to promote honesty\)](#).



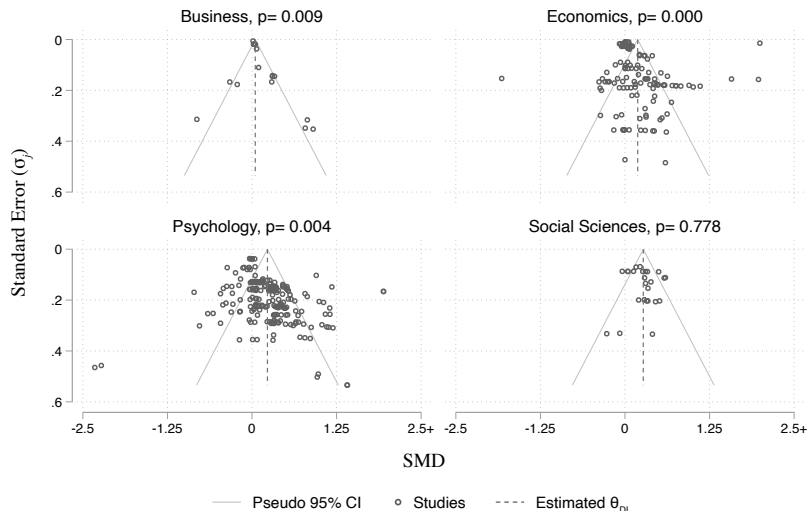
- **Prediction interval (95%):** $[-0.975; 1.457]$.
- Non-significant for 54.3% of studies; $> 0^{**}$ for 33.4% of them; $< 0^{**}$ for 12.2%.

Treatment effect as a function of take-up in the control



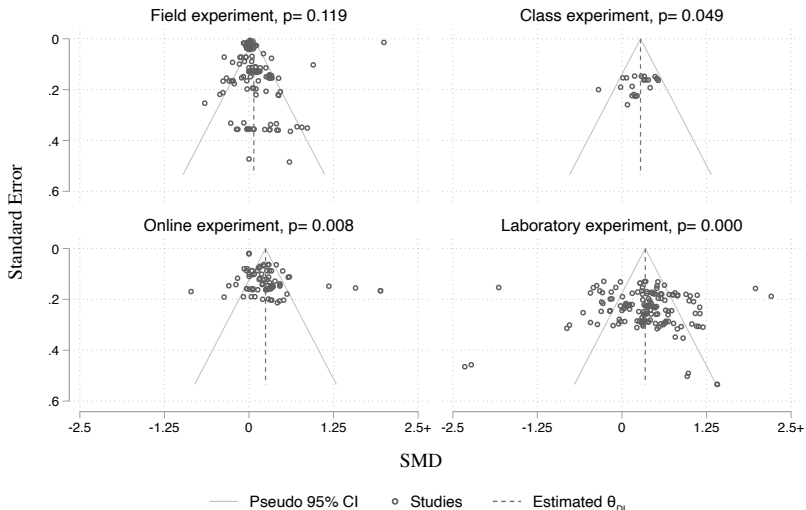
On average, $\Delta y = 6.8$ pp (from 42.7% in baseline) i.e., a 15.8% increase (computed on the 175 studies for which behavior can be normalized between 0 and 1).

How reliable are published results?... across disciplines



In parenthesis: p -value of the Egger statistic of small-study bias (H_0 : no relationship between SMD and s.d).

... across implementations



Results:

Methodological moderators

Nature of the comparison and elicitation method

Elicitation method

Field experiment	151		0.059 [-0.047, 0.165]	0.274
Class experiment	24		0.274 [0.191, 0.357]	0.000
Online experiment	82		0.245 [0.188, 0.302]	0.000
Laboratory experiment	158		0.422 [0.307, 0.537]	0.000
Test of group differences: $Q_b(3) = 21.41$, $p = 0.00$				

Incentives

No incentive	258		0.140 [0.059, 0.221]	0.001
Incentivised	157		0.378 [0.312, 0.445]	0.000
Test of group differences: $Q_b(1) = 19.75$, $p = 0.00$				

Results:

Heterogeneity in the effect

The nature of pro-social behavior

Target measure of pro-sociality

Study	K		Hedges's g with 95% CI	p-value
Intentions	35		0.040 [-0.007, 0.087]	0.093
Behavior	306		0.240 [0.165, 0.315]	0.000
Judgment	74		0.330 [0.235, 0.425]	0.000

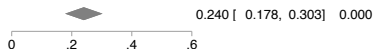
Test of group differences: $Q_b(2) = 39.27$, $p = 0.00$

Source of the reminder

Experimenter	253		0.265 [0.164, 0.365]	0.000
External authority	106		0.181 [0.140, 0.222]	0.000
Other participants	11		0.138 [-0.105, 0.381]	0.264
Personal	38		0.185 [0.099, 0.270]	0.000
Religious authority	7		0.109 [0.022, 0.196]	0.014

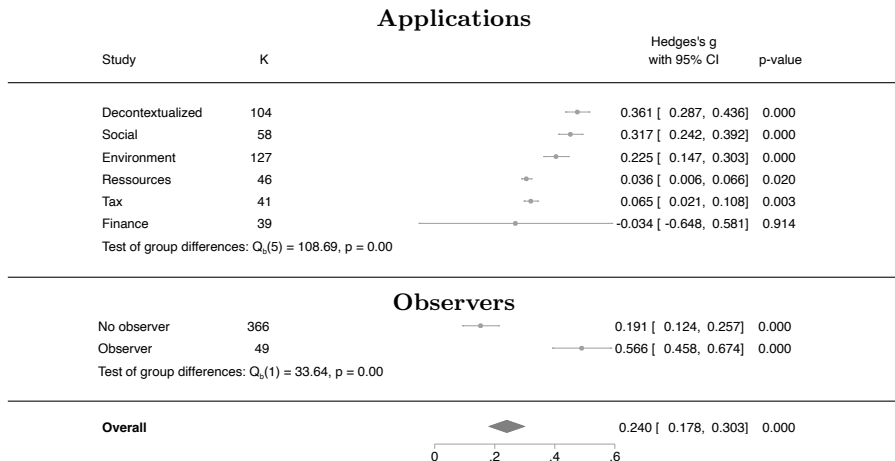
Test of group differences: $Q_b(4) = 5.39$, $p = 0.25$

Overall



- Effect on judgments > behavior > intentions.
- Stronger when the experimenter is the source (demand effect), but highly positive (even) otherwise.

Heterogeneity across applications



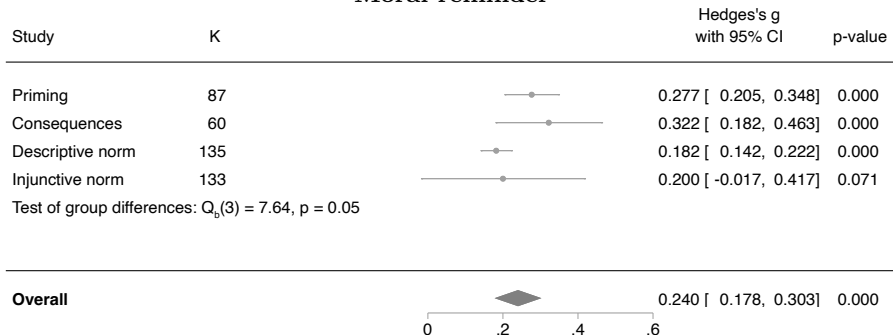
- Resource consumption and monetary applications undermine the effect.
- Explicit social image magnifies the effect.

Results:

Effectiveness of different kinds of interventions

How do MR compare to information provision?

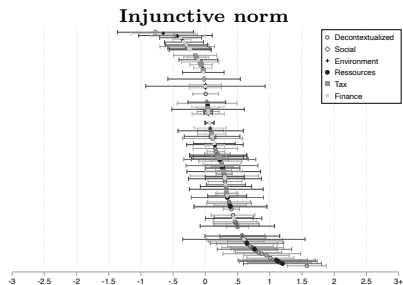
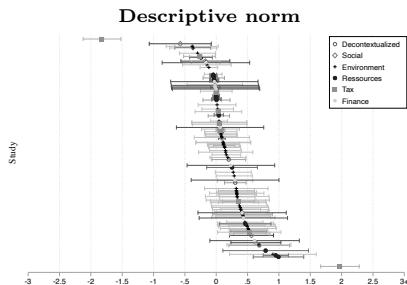
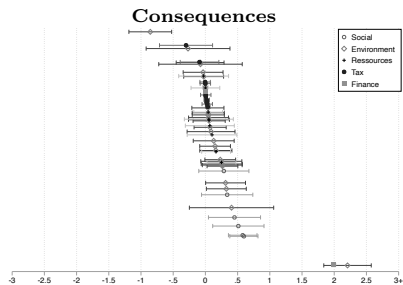
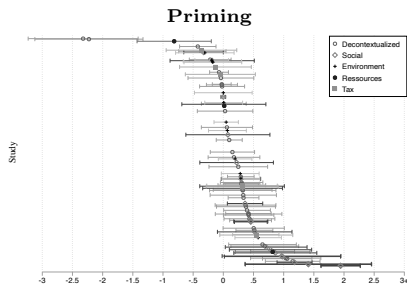
Moral reminder



Prediction intervals (also see [funnel plots](#)):

Priming:	[−0.233; 0.798]
Consequences:	[−0.747; 1.391]
Descriptive norm:	[−0.180; 0.544]
Injunctive norm:	[−2.280; 2.681]

Distribution of SMD per MR



Effectiveness of the tools: statistical evidence

	(1)	(2)	(3)	(4)
Consequences	0.075 (0.529)	0.164 (0.214)	0.325** (0.027)	0.436** (0.017)
Descriptive norm	-0.055 (0.572)	-0.053 (0.609)	0.057 (0.606)	0.148 (0.299)
Injunctive norm	-0.084 (0.395)	-0.072 (0.518)	-0.040 (0.734)	-0.026 (0.849)
Observer				0.302* (0.085)
Controls		✓	✓	✓
Methodological moderators			✓	✓
Implementation moderators				✓
Observations	415	415	415	415

Controls: discipline, geography, year

Methodological moderators: placebo, elicitation method, incentivized

Implementation moderators: observers, source, application

(details of variables used as [controls](#), [moderators](#), [sources of heterogeneity](#))

When moral reminders backfire

Predictors of a (significantly) negative effect

	(1)	(2)	(3)	(4)
Consequences	-0.037 (0.203)	-0.025 (0.437)	-0.038 (0.221)	-0.025 (0.595)
Descriptive norm	-0.002 (0.959)	0.002 (0.952)	-0.004 (0.901)	0.001 (0.975)
Injunctive norm	0.030 (0.371)	0.042 (0.245)	0.038 (0.216)	0.058* (0.090)
Observer				-0.079* (0.052)
Controls		✓	✓	✓
Methodological moderators			✓	✓
Implementation moderators				✓
Observations	415	415	415	415

Controls: discipline, geography, year

Methodological moderators: placebo, elicitation method, incentivized

Implementation moderators: observers, source, application

Conclusion

Conclusion

Provide an overview of 400+ different studies from 80+ articles on the effectivenesses of moral reminders as a tool to foster pro-social behavior.

- Large-scope, policy-oriented meta-analysis;
- Covering a wide range of different tools, behaviors, applications, and design choices.

Main conclusions:

- ① Moral reminders **do work** ! Small, but non-negligible, in magnitude (20% s.d. increase).
- ② Very heterogeneously though, regarding application and nature of pro-sociality: **highly context dependent, sometimes backfire**.
- ③ Information on consequences works just as well as priming; **norm interventions quite weak**.

Also highlights **the limitation of what can be learned from past accumulated evidence** ... and the need for comprehensive empirical studies generating the necessary exogenous variations in the main sources of heterogeneity.

ShinyApp website



<https://so-self.shinyapps.io/metaremindes/>

Snapshot

Thank you for your attention!

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Appendix

Keywords for inclusion

Extracted from references and secondary references of [Bénabou et al. \(2020\)](#).

Moral identity	Non-binding	Morality
Moral suasion	recommendation	Tax compliance
Moral licensing	Threatened morality	Social norms
Moral framing	Normative information	Moral hypocrisy
Moral self	Moral purity	Fairness
Moral compensation	Moral standards	Cooperation
Moral credentials	Conscience appeal	Honest
Moral hypocrisy	Recommended	Moral
Licensing effect	contribution	Social
Moral regulation	Moral persuasion	Ethic
Normative appeal	Experiment	Self
Moral appearance	Motivation	Norms
Moral self-worth	Framing	Behavior
Moral appeal	Public goods	
Moral cleansing	Self-concept	

Papers included

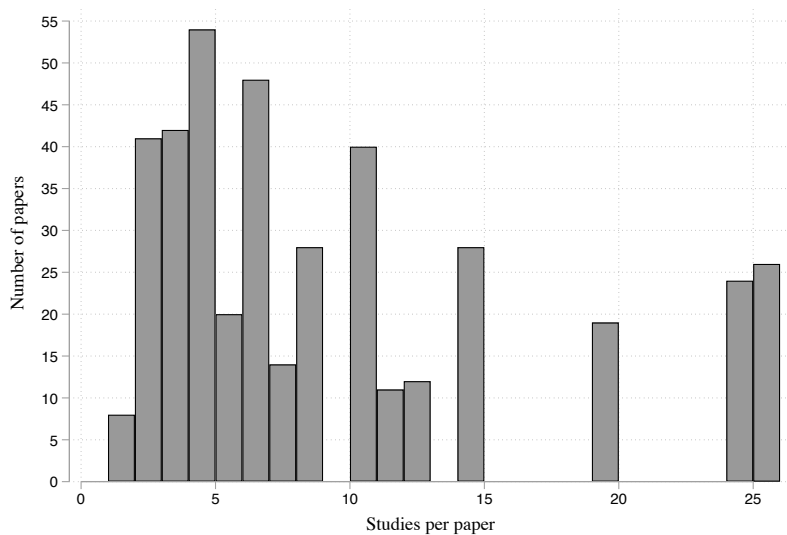
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Assumptions necessary to include MD studies

Assumption	# studies with group sizes	# studies without sizes
DiD comparisons	48	
Missing group sizes		45
Missing standard deviations		
inferred from standard errors	18	1
filled with p-values	8	17
filled with p-values' upper bounds	11	19
with p-values' UB & missing coeff.	5	2

[\(back\)](#)

Density of studies per research paper

[\(back\)](#)

Measure of SMD: Hedge's g

For each study, $k = 1, \dots, K$, MD provides:

- average outcome in the treatment ($\hat{y}_{kT}$) and control ($\hat{y}_{kC}$) groups;
- standard deviations ($\hat{\sigma}_{kC}^2$, $\hat{\sigma}_{kT}^2$);
- number of observations ($N_k = N_{kT} + N_{kC}$).

The Hedge's g corrects the small sample bias in the Cohen's d (see, e.g., [Schwarzer et al., 2015](#)):

$$\hat{\theta}_k = \left(1 - \frac{3}{4N_k}\right) \frac{\hat{y}_{kT} - \hat{y}_{kC}}{\sqrt{[(N_{kT} - 1)\hat{\sigma}_{kT}^2 + (N_{kC} - 1)\hat{\sigma}_{kC}^2] / (N_k - 2)}}$$

whose standard deviation is given by:

$$\hat{\sigma}_k^2 = \frac{N_k}{N_{kT} \cdot N_{kC}} + \frac{\hat{\theta}_k^2}{2(N_k - 3.94)}$$

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Random-effect model

Fixed-effect model: all SMDs are estimates of the same underlying true effect, θ .

- (obs.) Heterogeneity introduced through a (linear) function of p variables, $\mathbf{x}_k$, and unknown parameters $\mathbf{b}$:

$$\hat{\theta}_k = \theta_0 + \mathbf{x}_k \mathbf{b} + \varepsilon_k, \forall k, \text{ with } \varepsilon \approx \mathcal{N}(0, \hat{\sigma}_k^2)$$

- Can be estimated using the usual GLS estimator (weighted least square estimator with weights $\tilde{w}_k = 1/\hat{\sigma}_k^2$).

The random-effect specification allows for unobserved heterogeneity across studies:

$$\hat{\theta}_k = \theta_0 + \mathbf{x}_k \mathbf{b} + u_k + \varepsilon_k, \forall k, \text{ with } \varepsilon \approx \mathcal{N}(0, \hat{\sigma}_k^2), u_k \approx \mathcal{N}(0, \tau^2)$$

where $\hat{\tau}$ is the between study variance in the true effect.

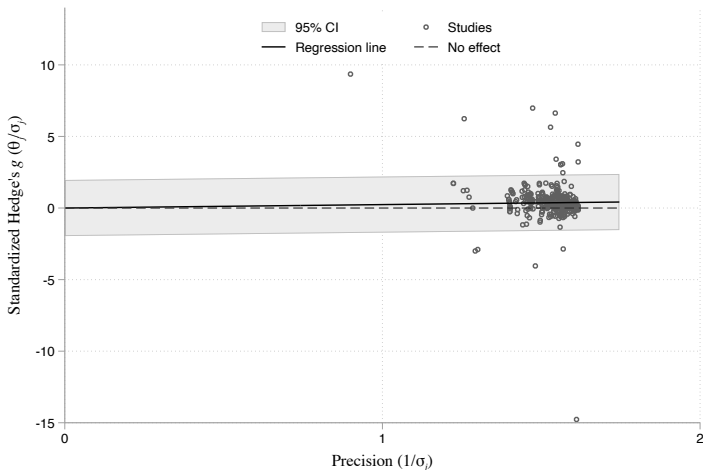
- GLS performed with weights $\tilde{w}_k = 1/(\hat{\sigma}_k^2 + \hat{\tau}^2)$.

The proportion of true variance in the effect is measured by:

$$l^2 = \frac{\hat{\tau}^2}{\hat{\sigma}_k^2 + \hat{\tau}^2}$$

How precise are the effects? Galbraith plot

Non-significant for 54.3% of studies; $> 0^{**}$ for 33.4% of them; $< 0^{**}$ for 12.2%.



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Descriptive statistics: methodological moderators

	Consequences	Priming	Descriptive norm	Injunctive norm	Overall
Elicitation method (%)					
Experimental game	13.3	60.9	32.6	45.6	40.0
Field task	53.3	11.5	45.2	36.8	36.6
Scale	33.3	27.6	22.2	17.6	23.4
Implementation					
Field experiment	51.7	13.8	43.0	36.8	36.1
Class experiment	0.0	5.7	10.4	3.7	5.7
Online experiment	38.3	20.7	13.3	19.1	20.3
Laboratory experiment	10.0	59.8	33.3	40.4	37.8
Incentives	21.7	50.6	27.4	46.3	37.6
Observer	1.7	12.6	0.0	27.2	11.7
Measure of the treatment effect					
Control	60.0	34.5	57.0	64.7	55.3
Placebo	40.0	65.5	43.0	35.3	44.7

Descriptive statistics: heterogeneity in the interventions¹

- **Target measure of pro-sociality** records the nature of the measure of pro-sociality targeted by the moral reminder (following [Ellemers et al., 2019](#)): Behavior (actual or self-reported) / Intention / Judgment.
- **Source** of the moral insight refers to who emits the moral reminder.

	Consequences	Priming	Descriptive norm	Injunctive norm	Overall
Target outcome (%)					
Intentions	0.0	20.7	11.1	1.5	8.4
Behavior	66.7	70.1	72.6	78.7	73.2
Judgment	33.3	9.2	16.3	19.9	18.4
Source of the reminder (%)					
Experimenter	50.0	55.2	59.3	69.9	60.5
External authority	50.0	9.2	37.0	13.2	25.4
Other participants	0.0	1.1	3.7	5.9	3.3
Personal	0.0	32.2	0.0	7.4	9.1
Religious authority	0.0	2.3	0.0	3.7	1.7
Observer (%)	0.0	13.4	0.0	25.7	14.2

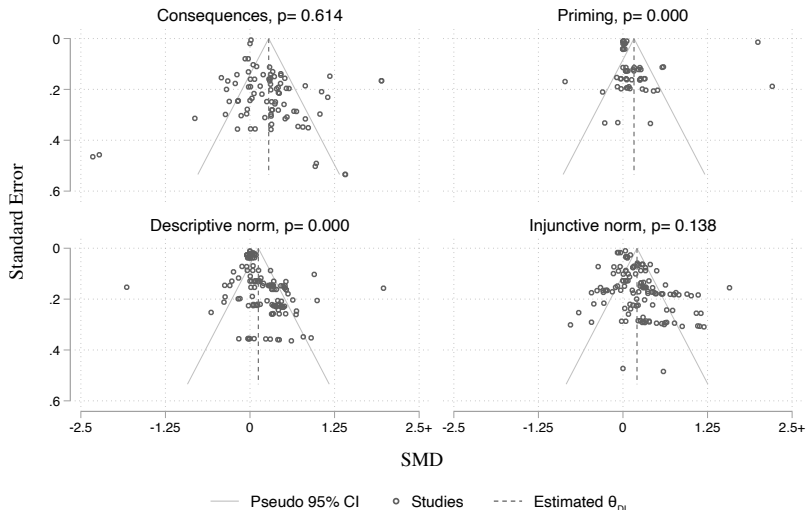
¹ All inputted variables have been classified independently by two RAs and cross-validated.

Descriptive statistics: heterogeneity in response

- **Application:** The policy sphere to which the target refers (“Social” is broad and refers to e.g., health, politics, etc.)

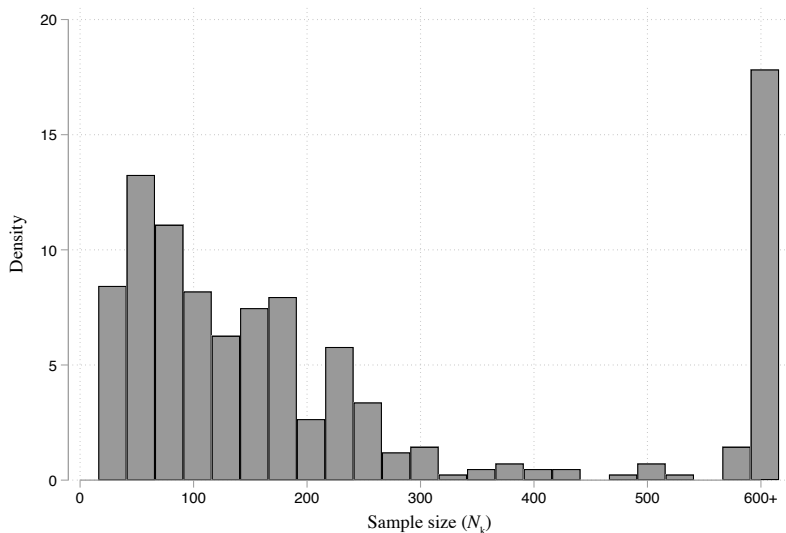
	Consequences	Priming	Descriptive norm	Injunctive norm	Overall
Application (%)					
Decontextualized	48.3	3.3	5.9	39.1	25.1
Social	26.4	11.7	7.4	13.5	14.0
Environment	14.9	45.0	54.1	10.5	30.6
Ressources	3.4	18.3	14.8	9.0	11.1
Tax	6.9	18.3	8.9	9.0	9.9
Finance	0.0	3.3	8.9	18.8	9.4
Share of females	44.6	56.4	52.0	49.3	52.0

Funnel plots, by tools



In parenthesis: p -value of the Egger statistic of small-study bias (H_0 : no relationship between SMD and s.d)

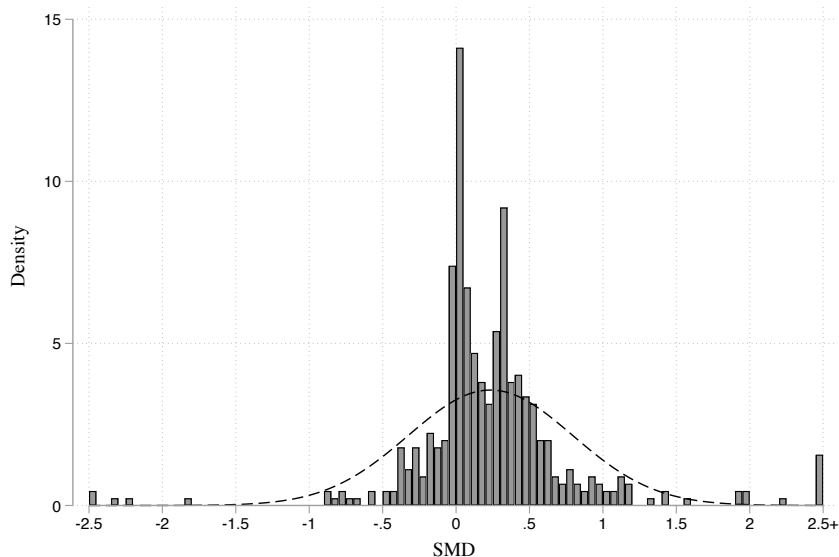
Distribution of sample size

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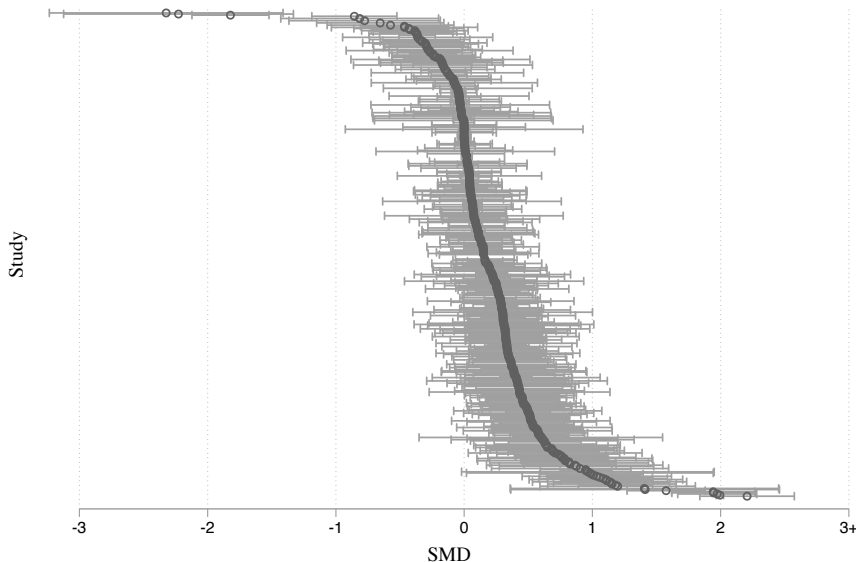
How effective are MR according to included studies?

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Effect size (SMD) measured as Hedge's g ([definition](#)) at the study level.



Extent of heterogeneity : direction of the effect

[\(back\)](#)

- Significantly positive studies: $\widehat{SMD} = 0.702 \in [0.580; 0.823], p = 0.000$.

ShinyApp website

